MME 356 - Marine Electrical Technology and Electrical Measurements

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E.A. Frimpong (PhD), MIEEE, MGhIE

Professor, Department of Electrical and Electronic Engineering,
Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

Course aim

- To provide insight into electrical propulsion system.
- Give an understanding of how maintenance of electrical equipment is done.
- Explain the basic concepts on measuring instruments.
- Explain the classification of requirements for electrical equipment for steering gears, electrical control from wheelhouse for electro-hydraulic steering gear.

Course Content

- **Essential equipment and special circuits:** Classification of requirements for electrical equipment for steering gears, electrical control from wheelhouse for electro-hydraulic steering gear, all electrical steering gear circuit and control equipment, navigation lights. Miscellaneous marine equipment and alarm system-electrical engine room telegraph, remote indicating revolution counter, remote helm indicator, salinity indicators, mist detectors, carbon di-oxide recorder, electrical equipment for watertight door operation, centralized alarm system in machinery space, for the above, fire alarm system for engine room, holds and accommodation, high temperature alarms low- and high-level alarm.

Course Content

- **Electric propulsion:** systems, differences, and relative merits, alternating current, diesel electric and turbo electric propulsion system engines, excitation system and control.
- **Maintenance of electrical equipment:** detection of faults and repair, preventive maintenance and periodic survey of equipment. Localization of cable faults. Classification of requirements for spares on board.

Course Content

- **Basics of measuring instruments** - deflection, control and damping devices moving coil, moving iron, dynamometer and thermocouple type of ammeter, voltmeter and wattmeter-their construction and other details, extension of scales of a meter. Single phase and three phase measurements by wattmeter. Measurement of energy. Measurement of speed, frequency and phase difference. Measurement of resistance, inductance and capacitance by Bridge Method. Magnetic measurement. Transducers and its application in the measurement of pressure, flow, temperature etc. Simple electronics measuring devices, such as VTVM, CRO, IC tester, signal generator.

Assessment

- **Continuous assessment** (Assignments, quizzes, mid-sem) - 30%
- **End-of-Semester Examination** (Written paper) - 70%

MAINTENANCE OF ELECTRICAL EQUIPMENT

Maintenance of electrical equipment

- Wear and tear in equipment is unavoidable.
- The life of equipment depends largely on how it is used and maintained.
- Maintenance of equipment keep them in running condition for long periods.
- All maintenance works done on ships must conform to the International Safety Management
- Code or ISM Code for Ships.

Maintenance of electrical equipment

- **Benefits of maintenance**
  - ✓ **Reduced downtimes**
  - ✓ **Reduced maintenance costs**
  - ✓ **Improved equipment lifespan**
  - ✓ **Reduced workplace disasters**
  - ✓ **Improved workplace culture**
  - ✓ **Improved workplace culture**

Maintenance of electrical equipment

- **Types of maintenance programmes**
  - ✓ **Breakdown or corrective maintenance**
  - ✓ **Preventive maintenance**
    - **Planned maintenance (resources and tasks)**
    - **Scheduled maintenance (who)**
  - ✓ **Condition maintenance system**
  - ✓ **Predictive maintenance**

Maintenance of electrical equipment

- **Benefits of predictive maintenance**
  - ✓ **Ability to schedule corrective maintenance in advance**
  - ✓ **It provides knowledge about parts to stock in advance**
  - ✓ **Reduced overhead cost**
  - ✓ **Energy savings**
  - ✓ **Overall efficiency improvements**
  - ✓ **Reduced maintenance and repair costs**
  - ✓ **Improved productivity**
  - ✓ **Fewer component failures**

Detection of faults

- Machinery various signs and symptoms that indicate fault conditions, before eventual failure.
- It is important for marine engineers to be able to identify faulty machinery and fix them before breakdowns occur.
- The following signs and symptoms can help:
  - Sound
  - Abnormal smell
  - High vibration
  - Leakages
  - Smoke
  - Abnormal parameters

Detection of faults

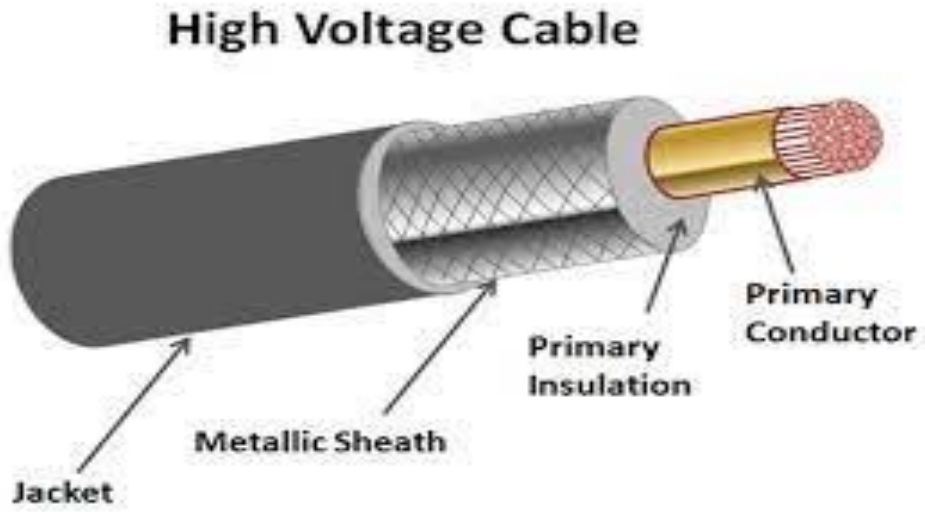
- Alarms
- Observing problems in connected systems
- Change in amperage

Spares onboard

- A sudden operational failure of critical equipment or technical systems onboard may result in a hazardous situation.
- Spares should be readily available to address such unforeseen problems
- A list of spares for critical equipment must be made, approved, and stocks provided before ships start sailing.
- Critical spares can be used either proactively or correctively.

Localization of cable faults

- Cable faults are difficult to locate.
- Types of cable faults include the following:
  - Short-circuits
  - Earth leakages
  - Cable sheath faults
  - Intermittent faults
  - Cable breaks



Localization of cable faults

- **The following approaches are used to locate cable faults:**
  - **Fault analysis**
  - **Pre-location**
  - **Tracing and pin-pointing**
  - **Cable identification**